

F65 65Aero Engine Runtime and Technical Supplement v1

Low-Level Runtime, Toolchain, Integration, and Evidence Constitution

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Status: APPROVED CANDIDATE DESIGN - NOT FINAL

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Platform: MEGA65 - 45GS02 + VIC-IV

Engine: 65Aero

Primary purpose: Complete low-level backend engineering umbrella for F-65 Megawing

Controlling parent: F65_Main_Concept_v1.6_FINAL_HUMAN_REVIEWED.md

Controlling-parent SHA-256:

e7d8ed40ce630d82e707e2a9c7f29995fac6f4281849c2c7ef5261d420c2c425

Player-facing companion: F65_Gameplay_and_Simulation_Supplement_v1_FINAL_HUMAN_REVIEWED.md

Gameplay SHA-256:

b675daa213c02e4b642fd8f8dd439c74f72808488c548a6af14fc8400487645a

Supersedes on approval: F-65 Engine Runtime and Toolchain Design Supplement Draft 0.2

Architecture effect: None. This document implements the approved Main Concept and Gameplay contracts and cannot relax either one.

Approval effect: The Runtime v1 prose constitution is human-approved as the candidate design. It is not FINAL and does not freeze machine layouts. FINAL / HUMAN-REVIEWED status requires the complete generated freeze package in Section 23, including exact approved-spec identity, generated interfaces, ABI, memory ledger closure, tests, and evidence artifacts.

Executive disposition

Engine Runtime Draft 0.2 remains useful as engineering provenance, but it is no longer suitable as the active implementation umbrella without a controlled rewrite. Its core ideas remain sound: C-primary target code, selective 45GS02 assembly, Java host oracles and generators, a narrow generated platform ABI, fixed pools, deterministic stages, declarative mission tooling, reproducible builds, evidence-first acceptance, and repository-scoped AI-assisted development.

Version 1 must, however, replace the Draft 0.2 authority model and pre-R0 assumptions with the frozen Revision 1.6 architecture, frozen Gameplay v1 behavior, completed R0-A/B/C evidence posture, current R0-D/E/F program, the two-domain SensorAndTrack contract, the current FCS state model, pro-

tected presentation/world-age model, final audio priority model, and the human-reviewed AI architecture.

This candidate therefore does **not** perform a cosmetic redline of Draft 0.2. It carries forward its valid engineering structure while replacing obsolete parent references, stale provisional memory assumptions, stale R0 sequencing, and placeholder subsystem boundaries.

The most important freeze condition is explicit: **Runtime v1 is not complete merely because this prose is approved.** The exact public layouts, memory subledgers, enum encodings, queue capacities, ABI records, build lock, acceptance catalog, and evidence schema required below must exist as generated, version-pinned repository artifacts and must agree with this document before Version 1 can be labeled FINAL / HUMAN-REVIEWED.

0. Purpose, authority, and active hierarchy

0.1 Purpose

This document defines how 65Aero implements the frozen Main Concept and Gameplay contracts on the MEGA65. It owns the low-level integration contracts that are intentionally absent from the player-facing specification, including:

- CoreRuntime ownership and module boundaries;
- the target compiler/toolchain model and C/45GS02 ABI;
- memory, MAP, DMA, IRQ, platform-service, stack, and code ledgers;
- deterministic tick dispatch, commands, events, entity lifecycle, and fixed-pool semantics;
- generated public interfaces and numeric registries;
- exact implementation ownership for input, presentation extraction, resources, packages, storage, replay, diagnostics, and evidence;
- integration interfaces for Flight Physics, Graphics, Audio, Radar/Sensors/Tracks, and AI;
- AircraftPhysicsClass representation and mission-load selection;
- R0-D/E/F measurement closure and the measured-limits handoff;
- Phase-1 integrated engine-harness architecture and acceptance;
- downstream Phase-2 through Phase-5 plug-in boundaries.

It deliberately does **not** duplicate the white papers' coefficient tables, doctrine weights, scan schedules, audio synthesis tables, renderer algorithms, or other subsystem depth when the integration contract can instead reference the controlling detailed paper and generated interface.

0.2 Active design hierarchy

The active design hierarchy is:

1. **F65 Main Concept v1.6** - master product and architecture authority.
2. **F65 Gameplay and Simulation Supplement v1** - player-visible and mission-visible behavior.
3. **F65 65Aero Engine Runtime and Technical Supplement v1** - this low-level implementation and integration umbrella after approval.
4. **Flight Physics and Simulation Engineering White Paper v3.3** - detailed physics/FCS/aircraft dynamics engineering.
5. **Graphics Engineering White Paper v2.1** - detailed graphics/world-registration/presentation engineering to the extent adopted by the frozen parents and not superseded by measured closure.
6. **Audio, Sound Effects and Music Engineering White Paper v1.0** - detailed audio engineering.
7. **Radar, Sensors and Track Engineering White Paper v1.0** - detailed SensorAndTrack engineering.
8. **AI Behavior and Decision Architecture White Paper v1.0** - detailed AI engineering.

Generated schemas, registries, ledgers, manifests, build locks, tests, and evidence records implement and prove this hierarchy. They are not independent design authorities.

0.3 Precedence

When two current sources touch the same subject:

1. Main Concept v1.6 controls architecture, ownership, determinism, clocks, memory boundaries, capacities, and phase gates.
2. Gameplay v1 controls player-visible behavior inside those invariants.
3. Runtime v1 controls implementation architecture inside both frozen parents.
4. Detailed white papers control subsystem engineering only where consistent with the core three.
5. Generated artifacts carry the exact machine representation of an already-approved contract; they cannot invent a new contract.
6. Code, tests, Xemu output, and hardware measurements are implementation/evidence and cannot silently rewrite specification authority.

A contradiction is escalated. It is not resolved by whichever file was edited most recently.

0.4 Retired sources

The following are retained for provenance but are not active authority after Runtime v1 approval:

- Architecture Revision 1.5.1 and earlier;
- Gameplay 0.2 and earlier;
- Engine Runtime Draft 0.2 and earlier;

- Read-First / Technical Alignment documents;
- F65-GAD-001 as a separate active decision layer;
- superseded R0 planning documents, except as evidence provenance for a closed gate.

No new overlay, first-read correction authority, or parallel architecture layer is created by this document.

0.5 Current development baseline

The current program state is:

- R0-A - **COMPLETE**;
- R0-B - **COMPLETE**;
- R0-C - **COMPLETE**;
- R0-D - **CURRENT**;
- R0-E - planned combined Xemu integration;
- R0-F - planned physical-MEGA65 confirmation;
- measured-limits closure - after accepted R0-F evidence;
- Phase 1 - integrated production-shaped 65Aero engine harness;
- Phase 2 - F-65 flight and aircraft systems;
- Phase 3 - radar, weapons, countermeasures, and damage;
- Phase 4 - tactical AI, Technical Combat Slice, and Midnight Spear;
- Phase 5 - campaign, compatibility, optimization, and release.

Runtime v1 does not reopen R0-A/B/C. New requirements from later human-reviewed subsystem papers are integrated through R0-D/E/F and the later production phases unless a genuine contradiction forces a recorded change.

1. Requirement classes and implementation discipline

Runtime v1 uses four requirement classes:

Class	Meaning
MUST	Durable implementation contract required by the frozen parent specifications.
TARGET	Engineering or quality target that may move through recorded evidence and human review.
MEASURED	Hardware/toolchain/performance/layout value frozen only through the named evidence path.

Class	Meaning
TBD	Required value, record field, table, threshold, capacity detail, or content decision whose owner/gate is named but not yet closed.

Rules:

- A TARGET, MEASURED, or TBD value cannot weaken a MUST.
- Prototype convenience values are never promoted to shipping constants merely because they compile or pass an emulator test.
- A measured value must carry build identity, configuration identity, method, and evidence provenance.
- When a field must exist semantically but its width is not yet approved, the logical field is specified here and its width remains generated/MEASURED.
- Generated files are never hand-edited when a source schema/generator owns them.
- Exact byte layout is accepted only when the generated source, C binding, Java binding/reference representation, static assertions, memory ledger, and test evidence agree.
- The 32 KB measured-limits reserve is not ordinary feature space and may not be consumed to hide an owner-ledger failure.

2. Target implementation, toolchain, and PlatformABI

2.1 Production language policy

Target runtime code is **C-primary** and compiled with the verified LLVM-MOS/MEGA65 toolchain established by the accepted R0 evidence.

Handwritten 45GS02 assembly is selective, first-class target code only when one or more of the following is true:

- the operation is hardware-facing and belongs behind PlatformABI;
- generated ABI glue requires it;
- measurement shows a material cycle/code-size benefit that is necessary to satisfy an accepted budget;
- an exact platform primitive cannot be expressed safely or predictably through the approved C subset;
- the implementation is explicitly part of a proven low-level primitive.

There is no fixed C/assembly percentage.

C is not a prototype language and assembly is not the default simply because the CPU is 8-bit lineage.

2.2 Host engineering language

Java remains the preferred independent host language for:

- high-precision physics oracles;
- bit-exact target reference models;
- interface/schema generation;
- numeric-registry generation;
- mission/package/asset compilation;
- golden-vector generation;
- evidence parsing and validation;
- offline analysis.

Host and target implementations must retain enough independence that one does not merely reproduce the same defect in two languages.

2.3 Deterministic target-C profile

Authoritative target code follows a restricted deterministic profile:

- no general-purpose authoritative-runtime heap;
- no uncontrolled recursion;
- no unbounded automatic allocation;
- no target floating-point arithmetic in authoritative simulation;
- public/deterministic/serialized/hardware-facing state uses fixed-width integer and generated fixed-point types;
- explicit range, rounding, saturation, overflow, and invalid-state behavior;
- bounded and instrumented stack use;
- no public ABI reliance on compiler-specific structure packing, bit-field layout, enum width, pointer width, or undefined behavior;
- no arbitrary MEGA65 physical address represented by an ordinary C pointer;
- no subsystem may execute MAP or start DMA hardware directly;
- no uncontrolled in-sortie code overlay;
- no runtime behavior may depend on host file-system ordering, nondeterministic iteration, or presentation frame rate.

2.4 Required build profiles

The repository shall provide at least:

Profile	Purpose
<code>host-test</code>	Java generators, schemas, high-precision/bit-exact oracles, deterministic fixtures, asset/mission/package tests.
<code>target-debug</code>	Bounds, canaries, assertions, fault injection, symbols, traces, stack high-water, maximum diagnostic visibility.
<code>target-profile</code>	Cycle, call, code/data/stack/DMA, latency and C-versus-assembly measurement using production-equivalent semantics.
<code>target-release</code>	Shipping configuration after all applicable gates and budgets pass.
<code>r0-proof</code>	Identified pre-Phase-1 proof configuration used only for the remaining R0-D/E/F closure work.

A single documented non-interactive build entry point shall, for the selected profile and scope:

1. validate the pinned specification/toolchain identity;
2. regenerate interfaces/registries/assets/mission fixtures in scope;
3. reject stale generated outputs;
4. run host tests and golden-vector tests;
5. compile and link the target;
6. create the identified D81/package image as applicable;
7. run supported Xemu smoke/regression tests when applicable;
8. emit maps, symbols, listings/disassembly where required;
9. emit memory/code/stack/runtime-support/DMA/cycle/latency evidence;
10. emit a machine-readable evidence index.

2.5 PlatformABI

C and handwritten assembly cross one narrow, generated, version-pinned ABI boundary.

Public PlatformABI entry points define and test:

- symbol and linkage rules;
- argument and return conventions;
- A/X/Y/Z/Q and processor-status preservation/clobber rules;
- compiler scratch/base-page obligations;
- hardware/software stack assumptions;
- IRQ/NMI assumptions;

- canonical MAP state;
- base-page restoration to the R0-established \$0200 game base page;
- \$01 / I/O personality requirements established by the accepted platform evidence;
- DMA-list lifetime and completion semantics;
- hardware-math wrapper semantics where used;
- video/audio hardware ownership;
- failure behavior and debug assertions.

The exact compiler, frontend, target, CPU-selection flags, linker flow, runtime/libc subset, assembler, Xemu configuration, MEGA65 core/ROM/system identity, and supported host environment are pinned in `toolchain/f65_toolchain.lock.json` or its approved successor. Runtime prose does not guess or silently update them.

3. CoreRuntime and module ownership

3.1 Engine module graph

The production-shaped 65Aero ownership graph is:

```
CoreRuntime
|-- PlatformABI
|-- InputEngine
|-- EnvironmentEngine
|-- ControlAndSystemsEngine
|-- FlightDynamicsEngine
|-- ContactEngine
|-- WeaponAndDamageEngine
|-- SensorAndTrackEngine
|-- AIEngine
|-- MissionEngine
|-- PresentationExtractor
|-- GraphicsEngine
|-- AudioEngine
|-- ResourceManager
|-- StorageService
`-- Diagnostics
```

A source directory layout may differ, but ownership cannot.

3.2 Sacred CoreRuntime boundary

CoreRuntime exclusively owns or arbitrates:

- the exact 100 Hz authoritative clock;

- tick dispatch and stage ordering;
- canonical mapping/base-page/stack/vector state;
- entity allocation and lifecycle commit;
- deterministic command and event ordering;
- global fixed-pool admission and owner-ledger validation;
- DMA arbitration/submission ownership through PlatformABI;
- RNG stream initialization and canonical deterministic-stream registry;
- authoritative checksum orchestration;
- presentation-extraction coordination and complete-snapshot publication;
- protected-service/deadline accounting;
- controlled performance-pause/fault behavior where an approved threshold requires it;
- build-visible instrumentation hooks needed to prove these contracts.

No gameplay or subsystem module may independently:

- alter the authoritative simulation clock;
- reorder a stage;
- execute MAP;
- start DMA hardware;
- publish a **PresentationSnapshot**;
- allocate outside its fixed declared pool/state owner;
- mutate another module's private arrays merely because the RAM is directly addressable.

3.3 Cross-module communication

Modules communicate through declared and generated interfaces only:

- read-only views;
- command frames;
- intent frames;
- deterministic events;
- core-owned bounded queues;
- fixed cross-tick mailbox/handoff state;
- typed entity/resource handles;
- immutable presentation fields.

Every public cross-module record has exactly one owner. Shared ownership is prohibited.

Every bounded queue or mailbox declares:

- owner;
- legal producers;
- legal consumers;
- production stage(s);
- consumption stage(s);
- maximum production fan-out;

- overflow/drop/reject behavior;
 - cross-tick lifetime;
 - source tick/generation or equivalent stale-data protection when needed;
 - checksum/replay participation if authoritative.
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4. Deterministic tick, causality, commands, and events

4.1 Single authoritative physical timeline

MUST: `ACTIVE_SORTIE` authoritative simulation runs at exactly **100 Hz**.

One authoritative tick is 10 ms. Every live authoritative physical aircraft, guided missile, countermeasure, gun group, ship/surface entity, and other physical entity advances on that same timeline according to its approved physical model.

There is no secondary physical-integration clock for distant, invisible, AI-controlled, or simplified entities.

KINEMATIC aircraft use a cheaper bounded physical model but remain owned by `FlightDynamicsEngine` and advance on the same 100 Hz physical timeline.

Lower-rate decision production is permitted only where explicitly scheduled. It never creates a second physical authority.

4.2 Exact 21-stage order

The complete authoritative tick order is:

1. Increment `SimulationTick`.
2. Latch `InputCommandFrame`.
3. Apply queued commands and eligible prior-tick directives/intents.
4. Update environment and carrier motion.
5. Resolve electrical, fuel, engine, and hydraulic supply.
6. Run flight-control laws and stability augmentation.
7. Apply actuator authority, rates, damage, asymmetry, and control mixing.
8. Sample atmosphere and calculate forces.
9. Integrate aircraft motion.
10. Resolve terrain, runway, deck, arrestment, and contact.
11. Accept weapon requests and create pending spawns.
12. Integrate existing weapons and countermeasures; consume eligible prior-tick sensor/support handoffs.
13. Detect collision, fuze, and damage events.
14. Accumulate and apply damage deterministically.
15. Run `SensorAndTrackEngine`.

16. Run scheduled RIO/AI decision services and publish bounded intents/commands for future eligibility.
17. Update mission objectives and scoring.
18. Commit despawns and spawns.
19. Extract bounded cockpit, warning, audio, and presentation state.
20. Calculate the canonical authoritative checksum.
21. Atomically publish a complete **PresentationSnapshot**.

No subsystem may pull a downstream effect earlier in the same tick for convenience.

4.3 Stage-15 / stage-16 causality

Stage 15 completes legal sensor/track state. Stage 16 AI/RIO may consume that completed state and produce **AIIntentFrame** and other approved outputs.

An output produced at stage 16 on tick N becomes eligible no earlier than the legal application point on tick N+1.

If a lower-cadence AI service is not due, its valid held intent persists while the owning 100 Hz controller continues to close against it every physical tick.

Stage-16 output cannot create same-tick feedback into:

- flight physics;
- FCS/system authority;
- weapon integration;
- sensors/tracks;
- damage;
- mission state.

4.4 Simultaneous-event ordering

Unless a more specific approved subsystem rule narrows the order without violating this one:

- deterministic events sort by event class, source handle, target handle, then producer sequence;
- commands valid at tick start execute before damage produced later that tick;
- mutual kills are legal;
- component damage accumulates before capability recalculation;
- freed slots are not reused before stage-18 lifecycle commit;
- new entities begin scheduled updates on the following tick;
- presentation ordering cannot change simulation ordering.

4.5 Command and event envelopes

The generated interface registry shall define at minimum:

- **CommandHeader** / command envelope;
- **EventHeader** / deterministic event envelope;
- producer sequence semantics;
- source/target handle representation where applicable;
- source tick/eligible tick fields where required;
- queue identity and maximum depth;
- reject/drop/fault behavior.

Command/event payloads are typed by generated IDs. Unknown or version-incompatible payloads fail deterministically; they are never interpreted by size guessing.

4.6 Pause

Full pause freezes authoritative simulation advancement and therefore **SimulationTick**.

UI/presentation work may continue, but pause cannot create new physical, sensor, weapon, damage, AI-authoritative, or mission events.

5. Generated public interfaces and record constitution

5.1 Single source of machine layout

The exact public binary contract is generated from machine-readable source, not copied independently into C headers, Java classes, assembly include files, and documentation.

At minimum the generated interface system shall produce:

- C declarations and static assertions;
- assembly constants/offsets;
- Java declarations/reference representations;
- enum/value registries;
- field offsets/sizes/alignments;
- owner and lifetime metadata;
- schema/version IDs;
- compatibility checks;
- memory-ledger charges.

The candidate logical record catalog below is normative in **meaning and ownership**. Exact width/alignment/encoding is normative only when already frozen by Main/Gameplay or when the generated interface package is approved.

5.2 Core/public record catalog

The generated catalog shall include, as applicable:

Core identity and memory

- **EntityHandle** - **exactly 4 bytes**: entity type, pool index, 16-bit generation. Zero/generation-zero forms invalid/reserved.
- **FarPtr32** or approved physical-address type.
- **ResourceHandle16** or approved resource-handle representation.
- address-class and DMA-request records.
- **SimulationTick** representation.
- deterministic RNG stream/state registry entries.

Input and control

- **InputCommandFrame**;
- semantic control-context fields;
- command-edge/repeat representation;
- **FCSMode**;
- **FCSStatus**;
- **ADLCState**;
- autothrottle state;
- pilot trim state;
- temporary FCS transfer-bias state where authoritative.

Aircraft physics and systems

- **AircraftPhysicsClass**;
- per-aircraft physical-model selection record;
- system capability/read-only projections used across modules;
- **ControlSurfaceState**/approved actuator-state handoff if public;
- carrier/contact references where cross-module.

Sensor/weapon handoff

- **SensorHandoffState**;
- **SeekerAssessment**;
- **WeaponSupportUpdate**;
- **LinkTrackReport**;
- tactical knowledge-domain identifier;
- **SensorTrack** / semantic tactical-track representation;
- RIO-priority selection representation;
- **SeekerInputView**;
- **CountermeasureView**;
- **EmitterView** where exposed;
- **RWRContactState**;

- `MissileThreatState`;
- own-weapon/read-only guidance projections required by `SensorAndTrack` and AI.

AI

- `AIIntentFrame`;
- `GuidanceIntent`;
- `SystemIntent`;
- `WeaponIntent`;
- `SensorIntent`;
- held-intent validity/lifetime representation;
- legal-knowledge view headers/versioning;
- `VisualContactView`;
- `OwnWeaponStatusView`;
- `WeaponEmploymentProjection`;
- `FacilityClearanceView`;
- `RecoveryReference`;
- `RoutePlan` and route-leg representation;
- `TaxiRoute`;
- `DeckRoute`;
- doctrine/profile references;
- suspended/resume route state where authoritative.

Presentation

- `PresentationSnapshot`;
- snapshot identity/generation/source-tick fields;
- `WorldRegistrationRecord`;
- world buffer/generation identity;
- displayed-world age source fields;
- cockpit/HUD/status/RIO display fields;
- warning/audio presentation event fields;
- jammer / organic-radar-blanked / MEGA-link-denied presentation state required by Gameplay v1.

Resource/package/storage/save

- package/chunk headers;
- resource manifest entries;
- asset records;
- mission compiled records;
- capacity witness;
- save header/version/migration/failure records;
- evidence index records.

5.3 Exact-layout freeze rule

Runtime v1 cannot be labeled FINAL until the Version-1 generated interface package:

1. includes every cross-module/cross-tick/public record required by the frozen parents and active detailed engineering;
2. has no duplicate independent source of field offsets or enum values;
3. compiles C and assembly bindings with static assertions;
4. produces matching Java representations/golden fixtures;
5. closes the applicable memory owner ledger;
6. survives replay/checksum and serialization tests where applicable;
7. is pinned by exact source/generator/output identity.

6. Memory, MAP, DMA, code, stack, and owner ledgers

6.1 High-level fast-memory ownership

The Main Concept fast-memory map remains the parent contract:

Physical range	Size	Primary ownership
\$000000-\$00FFFF	64 KB	Resident executable, base page, stack, vectors, scheduler, hot state
\$010000-\$017FFF	32 KB	Active simulation, entities, aircraft systems, SensorTrack, AI and tactical state
\$018000-\$01BFFF	16 KB	Audio state, mission-hot data, bounded queues and cross-tick handoff classes
\$01C000-\$01CFFF	4 KB	Display pointer/control records
\$01D000-\$01FFFF	12 KB	Shared scratch/transient CPU workspace
\$020000-\$02FFFF	64 KB	Display store A
\$030000-\$03FFFF	64 KB	Display store B

Physical range	Size	Primary ownership
\$040000-\$047FFF	32 KB	Cockpit/HUD/MFD/sprite/palette assets
\$048000-\$04FFFF	32 KB	Renderer clipping/bucket/occlusion/span workspace
\$050000-\$057FFF	32 KB	Resource staging and DMA lists
\$058000-\$05FFFF	32 KB	Measured-limits reserve

VIC-IV Color RAM and Attic RAM remain distinct memory resources with their parent-defined ownership. They are not general substitutes for active authoritative RAM.

6.2 Hard active-simulation closure

Before downstream resident-state admission, the generated owner ledger must prove the complete **\$010000-\$017FFF** active-simulation subledger.

At minimum the closure explicitly charges:

- entity common state and fixed pools resident in the region;
- F-65 systems state resident in the region;
- both capacity-isolated semantic SensorTrack domains;
- **SensorHandoffState** to the exact parent-mandated owner/subledger treatment;
- command/intent persistent state;
- AI authoritative state and held intents;
- alignment/padding;
- debug-guard policy as applicable;
- any cross-module record whose accepted generated placement falls in the region.

The old Draft-0.2 single-domain track allocation and generic AI-intent placeholder are not reusable as a proof.

The old single-24-track interpretation is superseded. Current semantics require two 24-track tactical knowledge domains, MEGA/BLUE and RED, with a maximum of 48 physical semantic-track records. The exact final track size remains generated/MEASURED.

The generated ledger shall use one unambiguous owner placement for every cross-track record. **Default Version-1 placement is that SensorHandoffState is owned and fully charged to SensorAndTrackEngine inside \$010000-\$017FFF.** This includes the seeker-assessment mailboxes, weapon-support-update mailboxes, and the tactical-link pending state that are members of that record. The

generic \$018000-\$01BFFF queue band may hold other bounded queues that are not members of `SensorHandoffState`; it is not a second place to split or hide the `SensorAndTrack` handoff charge. If the generated active-simulation ledger cannot close with this default, the record design or an explicitly approved allocation decision must change before downstream residency is admitted.

6.3 Reserve rule

\$058000-\$05FFFF is measured-limits reserve.

- It is not baseline feature capacity.
- It cannot be consumed to make an owner ledger appear to fit without a recorded design decision and updated generated allocation.
- A reserve draw, if ever approved, is explicit, measured, and traceable.

6.4 MemoryAccessABI and MAP

Only the protected CoreRuntime/Platform layer may manipulate MAP or arbitrary physical-memory windows.

Public physical-memory access uses approved typed physical addresses/handles and wrappers. MAP windows are temporary and restore the canonical CPU-visible state before public return or yield.

Critical resident code, vectors, stack, and base page cannot be hidden by a temporary mapping state.

6.5 DMA

Only CoreRuntime/Platform DMA services start DMA jobs.

Every DMA request declares:

- normalized source/destination physical address;
- source/destination address classes;
- byte length;
- overlap legality;
- list-storage lifetime;
- deadline/priority class;
- completion semantics;
- protected audio/display interaction;
- instrumentation identity.

A blocking hardware job is not described as preemptible merely because the software submission path is asynchronous.

World/presentation DMA is split or deferred as required to satisfy protected-service latency contracts.

6.6 Stack, code, and runtime-support ledgers

Every target build emits per-owner:

- code bytes;
- read-only data bytes;
- writable authoritative state bytes;
- transient/scratch bytes;
- stack high-water and guard result;
- compiler/runtime support pulled in by that owner when attributable;
- generated interface bytes;
- DMA-list/staging charge;
- reserve use;
- MEASURED limit and margin.

No average stack use or average code residency substitutes for worst legal path/high-water evidence.

7. Entity pools, lifecycle, and AircraftPhysicsClass

7.1 Fixed capacities

The current hard capacities are:

Pool/state	Hard capacity
Aircraft	16
Ships and carriers	8
Surface radar/SAM entities	16
Guided missiles	32
Gun projectile groups	32
Chaff/flare entities	64
Dynamic mission entities	32
Simulation-relevant effects	32
Presentation effects	64
Radar truth contacts	32 shared physical candidates
Semantic tactical tracks	24 per tactical-link knowledge domain
Active tactical knowledge domains	2 - MEGA/BLE and RED
Maximum physical semantic-track records	48
RIO priority tracks	4 - MEGA-domain subset only
Active objectives	16

Pools never grow dynamically.

7.2 Lifecycle

- allocation selects the lowest legal free pool index unless a more specific parent rule says otherwise;
- despawns and allocations occur only through the stage-18 lifecycle commit;
- spawn requests sort deterministically by class priority, requester, and sequence;
- freed slots are not reused before commit;
- new entities begin scheduled updates on the next tick;
- required-entity exhaustion is a mission/compiler/admission failure or a class-specific deterministic rejection, never silent replacement;
- droppable presentation effects follow presentation degradation policy and are counted.

7.3 AircraftPhysicsClass

Every aircraft instance has a mission-compiled `AircraftPhysicsClass` selected before the sortie becomes active.

The Version-1 semantic classes are:

- `SIX_DOF` - full approved aircraft physical model;
- `KINEMATIC` - bounded lower-cost physical model, still advanced at 100 Hz and still owned by `FlightDynamicsEngine`.

The exact enum encoding and record placement are generated.

Runtime may not silently change an aircraft's physics class during a sortie to recover performance.

If a later design deliberately permits a class transition, it requires parent approval, deterministic transition semantics, mission/compiler admission, replay behavior, and new acceptance evidence.

7.4 Required nine-aircraft harness

The baseline combined-load harness contains **9 simultaneously live aircraft**:

- 6 `SIX_DOF`;
- 3 `KINEMATIC`;
- the baseline AIC is `KINEMATIC`.

A `SIX_DOF` AIC is a separately identified compiler-proved/timing-evidence harness variant. It is not silently substituted into the baseline.

The mission compiler records the physics class for each aircraft and emits a capacity/timing witness.

7.5 Combined-load acceptance profile

The integrated overlap case includes, concurrently where legal:

- 9 aircraft;
- 16 guided missiles;
- 24 gun projectile groups;
- 48 live decoys;
- 8 dynamic mission entities;
- 8 active objectives;
- 64 presentation effects;
- required MEGA and RED track/domain work;
- damage/event work;
- AI/RIO scheduling fixtures or later production AI according to phase;
- protected HUD/cockpit/audio;
- outside-world construction/rendering.

This is a required overlap case, not authorization to assume all hard pool maxima coexist at once.

8. Performance constitution and instrumentation

8.1 Performance ownership

Runtime v1 owns the detailed performance constitution. Main and Gameplay define visible behavior and top-level invariants; Runtime owns the measured engineering ledgers that prove them.

The performance constitution includes:

- per-tick CPU clocks;
- per-stage clocks;
- protected-service clocks;
- DMA contribution and blocking intervals;
- IRQ service contribution;
- input event-to-latch latency;
- simulation-event-to-warning/audio latency;
- `PresentationSnapshot` publication latency;
- completed-world cadence;
- displayed-world age;
- view-change-to-first-complete-world timing;
- queue/pool high-water;
- code/data/stack high-water;
- reserve use;
- p95 and legal worst-case measurement where meaningful;
- worst legal phase alignment, not only averages.

8.2 Non-negotiable bars

- **ACTIVE_SORTIE** authoritative simulation remains exactly 100 Hz.
- A late world frame cannot slip, merge, stretch, or reorder a simulation tick.
- Protected cockpit/HUD composition, critical warnings, input service, protected audio, and other protected services retain their approved deadlines independently of world construction.
- World construction may yield and resume and publishes complete buffers only.
- Sustained completed-world cadence below **20 Hz** is a failure condition.
- Approximately **30 Hz** completed-world cadence in ordinary cruise is a TARGET.
- Approximately **25 Hz** completed-world cadence under the approved combined-combat load is a TARGET.
- Maximum displayed-world age is MEASURED through R0-D/E, confirmed on physical MEGA65 in R0-F, and frozen only through measured-limits closure.
- Presentation degradation sheds optional detail/cost before protected-service failure.
- **Never-shed contract:** overload handling may never skip, demote, or defer the authoritative 100 Hz simulation tick; the player's **SIX_DOF** physical update; protected HUD/cockpit composition; P0/P1 audio service; the legally scheduled seeker/support consumption point; or the canonical checksum stage. These are not quality knobs.
- The normal presentation shedding order reduces optional effects/detail and world LOD first, then may reduce completed-world cadence only within the approved presentation contract. Sustained world cadence below 20 Hz is a failure, not an additional degradation tier.
- If the legal load still cannot preserve the never-shed set, the mission/compiler/load ledger or measured configuration fails acceptance; the runtime does not invent another clock, demote a physics class, spend reserve silently, or skip authoritative work.
- Presentation cadence or LOD cannot change physics, sensor/weapon causality, AI knowledge, RNG, or authoritative checksum.

8.3 Historical 530,000-clock fixture

The historical **530,000-clock protected non-render workload** remains an R0-D calibration fixture because Main v1.6 explicitly carries it into Delta.

It is **not** a frozen shipping per-tick budget and does not automatically reproduce Draft-0.2's earlier planning partition.

Draft-0.2 values such as protected HUD/cockpit, world, or reserve planning allotments may be retained in provenance/evidence notes only if clearly labeled as historical planning fixtures. Runtime v1 production limits are set from accepted

R0-D/E/F and measured-limits evidence.

8.4 Required counters

At minimum R0-D/E/F instrumentation exposes machine-readable counters for:

Core/timing

- total authoritative tick clocks;
- each stage clocks;
- deadline debt;
- controlled timing faults;
- IRQ clocks;
- DMA clocks/blocking intervals;
- stack high-water.

World/graphics

- world construction start/complete generation;
- source `PresentationSnapshot` tick/generation;
- displayed world generation;
- displayed-world source tick and age;
- completed-world cadence;
- view-change-to-first-complete-frame latency;
- face/vertex/span/bucket/occlusion/registration-anchor/LOD high-water as applicable;
- dropped/shedded presentation work by class;
- graphics DMA contribution.

Audio

- event-to-audio latency;
- P0 worst latency;
- audio service clocks;
- SID-voice high-water;
- PCM-channel high-water;
- music voices stolen;
- PCM preemptions by priority;
- dropped effects by priority;
- speech interruptions;
- audio cache high-water;
- staging bytes/latency;
- DMA contention;
- underrun/distortion faults.

AI

- stage-16 scheduler clocks;
- per-service due/not-due counts;
- reflex/doctrine/mobility worst clocks;
- held-intent age/high-water;
- AI resident code/data/state bytes;
- route/doctrine mission-hot bytes;
- AI scratch/stack high-water;
- legal-knowledge-view bytes;
- intent/event production high-water.

Sensor/weapon

- domain track high-water;
- truth-contact high-water;
- link queue high-water/drop counts;
- SensorHandoff mailbox overwrite/supersession counts;
- seeker/support handoff age;
- stage-15 clocks by major logical service;
- weapon stage-12 handoff consumption counts/age.

No average measurement substitutes for the required worst legal phase evidence.

9. InputEngine and InputCommandFrame

9.1 Ownership

InputEngine owns hardware/input-device sampling and production of the bounded semantic **InputCommandFrame** consumed by stage 2.

Input does not directly mutate flight, sensor, weapon, mission, or presentation-owned private state.

9.2 Semantic contexts

The four Gameplay contexts remain:

- DECK;
- TFL - takeoff and landing;
- NORMAL_FLIGHT;
- COMBAT.

Context changes input semantics/presentation but does not itself select the physical FCS mode.

9.3 Required Version-1 behavior

The generated `InputCommandFrame` must support the approved player-visible controls without freezing a keyboard/joystick key map as architecture.

Version 1 keeps defensive countermeasure execution RIO-default according to Gameplay v1. If a later numbered Gameplay revision adds an optional manual CM action, the semantic input schema is extended through a compatible numbered interface change rather than an ad hoc raw-key path.

Entering Combat may capture autothrottle at current KIAS according to Gameplay behavior; Runtime must represent the state transition explicitly and must never silently recapture it during later maneuvering.

9.4 Latency evidence

R0-D/E/F and Phase-1 evidence record:

- physical/input event timestamp or approved test-source identity;
- `InputCommandFrame` latch tick;
- downstream command application tick;
- presentation feedback tick where relevant;
- p95/worst measured latency in the accepted configuration.

Exact ceilings are MEASURED until frozen by the evidence path.

10. EnvironmentEngine, ContactEngine, world queries, and carrier state

10.1 Environment ownership

`EnvironmentEngine` owns the authoritative environmental state and query interfaces required by physics, sensors, graphics registration, and mission logic, including approved atmosphere/wind/terrain/query representations.

Subsystems consume declared queries/views rather than maintaining divergent private terrain truth.

10.2 Visual/query registration

Graphics may render an older complete world buffer than the newest authoritative simulation tick. Runtime must therefore preserve explicit registration between:

- the authoritative query state used by physics/sensors/contact;
- the `PresentationSnapshot` from which a world buffer was constructed;
- the `WorldRegistrationRecord` associated with that displayed buffer.

Spatial visual cues tied to the displayed world must use the displayed buffer's registration generation. Newer textual/status information may be drawn from the current protected presentation path only where Gameplay/Graphics allow it.

HUD/cockpit does not disappear because outside-world construction is late.

10.3 Contact and carrier

ContactEngine owns terrain/runway/deck/arrestment/contact resolution at stage 10.

Carrier/deck coordinate and contact semantics remain those defined by the Main/Physics contracts. Carrier motion is authoritative and not derived from presentation interpolation.

Facility-clearance state used by AI is a bounded read-only view of owning facility/contact/mission state; AI does not directly acquire runway/catapult/arresting ownership by mutating facility internals.

11. FlightDynamicsEngine and ControlAndSystemsEngine integration

11.1 White-paper relationship

Physics v3.3 owns detailed equations, coefficient tables, mass/inertia schedules, FCS gains, actuator/hydraulic models, unaugmented envelope, atmosphere/engine details, and physics oracle acceptance.

Runtime v1 owns the integration boundary, data ownership, stage placement, generated public records, mission-load class selection, cycle/memory evidence, and fault/replay integration.

11.2 Current FCS state model

The implementation uses the separated parent concepts:

```
ControlContext = DECK / TFL / NORMAL_FLIGHT / COMBAT
FCSType        = AUGMENTED / AUG_OFF
FCSType        = NORMAL / DEGRADED / DIRECT_ONLY
ADLCState      = OFF / CAPTURE / TRACK / DEGRADED
ADLCLimitingReason = generated reason code, including an explicit FAILED reason when failure
Autothrottle   = separate state machine
```

FCSType records selected/commanded mode. **FCSType** records actual delivered augmentation capability. **DIRECT_ONLY** can force direct behavior without rewriting the pilot's switch state.

ADLCState keeps the Main Concept vocabulary OFF / CAPTURE / TRACK / DEGRADED. Gameplay's requirement to distinguish a failed ADLC is represented in the **same generated ADLC record** by ADCLimitingReason (or an approved equivalent reason field). A hard failure therefore presents as ADLCState = DEGRADED with a failure reason, rather than creating a competing second top-level ADLC vocabulary. Exact reason-code values and width are generated-interface work.

The old Assisted/Manual architecture is retired and cannot reappear in generated enums or code as a public mode alias.

11.3 Command chain

The physical command chain is conceptually:

```
InputCommandFrame
-> semantic control context
-> FCSMode + FCSStatus
-> optional ADLC overlay
-> commanded control quantities
-> non-augmenting mixer/gearing
-> hydraulic authority
-> actuator rates/stops/damage/asymmetry
-> ControlSurfaceState
-> FlightDynamicsEngine forces/moments/integration
```

AUG_OFF remains direct/manual flying, not a second hidden rate-command law. It retains pilot trim, non-augmenting mixer/gearing, hydraulics, actuator limits/rates, damage/asymmetry, and natural aerodynamic stability while removing closed-loop G/roll-rate command, AoA/G limiting, and artificial stability damping.

11.4 Bumpless transfer

A change in FCS capability/mode cannot create an artificial discontinuity merely because a switch/failure changed.

The generated state model includes any required bounded temporary transfer-bias state separately from persistent pilot trim. Transfer bias decays to zero and cannot silently become trim.

11.5 ADLC and autothrottle

ADLC is an approach-control overlay, not a peer fundamental flight law. Selecting AUG_OFF while ADLC is engaged disengages or degrades ADLC through the approved state transition before direct authority becomes active.

Autothrottle is independent. FCS mode changes do not rewrite autothrottle state except where a specific approved safety/state rule requires it.

11.6 KINEMATIC aircraft implementation gate

Runtime v1 defines the KINEMATIC interface boundary but does not invent its final approximation/tuning in place of the Physics/AI implementation gate.

The implementation must prove:

- 100 Hz advancement;
 - ownership by `FlightDynamicsEngine`;
 - bounded state;
 - deterministic update;
 - no alternate uncontrolled motion authority;
 - mission-compiled class selection;
 - replay/checksum participation;
 - acceptable handoff to contact/sensor/presentation systems;
 - measured code/state/cycle cost.
-

12. WeaponAndDamageEngine and SensorAndTrackEngine integration

12.1 Ownership boundary

`SensorAndTrackEngine` owns observation and assessment.

It owns, as applicable:

- radar/IR observables;
- organic radar observations;
- semantic tactical-track processing;
- MEGA LINK / RED LINK ingest/fusion/output;
- RWR emission assessment and threat-state inputs;
- chaff/flare competition against sensors/seekers;
- `SeekerAssessment` generation;
- `WeaponSupportUpdate` generation from valid local fire-control state.

`WeaponAndDamageEngine` owns:

- authoritative missile/guidance state;
- physical missile state and 3DOF evolution;
- support/autonomy transitions;
- target/lock/countermeasure state;
- guidance integration;
- fuzes/detonation;
- damage and weapon lifecycle;
- cannon/projectile-group physical authority.

SensorAndTrack may assess what a seeker can see. It cannot mutate missile guidance state directly.

12.2 SensorHandoffState

Cross-tick sensor/weapon/link handoff is fixed persistent state owned and byte-accounted by **SensorAndTrackEngine**.

It contains at minimum:

- 32 pending **SeekerAssessment** mailboxes - one per guided-missile slot;
- 32 pending **WeaponSupportUpdate** mailboxes - one per guided-missile slot;
- one bounded tactical-link pending queue with generated capacity/fan-out/drop semantics.

Exact field widths, alignment, padding, queue capacity, mailbox generation/supersession rules, and total bytes are generated/MEASURED.

No dynamic allocation is permitted.

Weapon stage 12 consumes eligible mailboxes read-only. Sensor stage 15 may overwrite/supersede only according to the generated source-tick/generation rule after the legal consumer boundary.

12.3 One-tick sensor/weapon causality

- Stage-11 weapon acceptance uses sensor/track state already legal at tick start.
- Stage-12 missile support cannot depend on an observation/report/handoff created at stage 15 of the same tick.
- A **WeaponSupportUpdate** produced at stage 15 on tick N becomes consumable no earlier than stage 12 of tick N+1.
- A **SeekerAssessment** produced at stage 15 on tick N becomes consumable no earlier than stage 12 of tick N+1.
- A tactical-link report generated on tick N becomes eligible for recipient ingestion no earlier than a later legal tick boundary.
- AI/RIO consuming stage-15 state produces commands/intents for no earlier than the following tick.

12.4 Tactical knowledge domains

There are exactly two current capacity-isolated tactical knowledge domains:

- MEGA / BLUE;
- RED.

Each domain has capacity for up to 24 semantic tactical tracks. Association, fusion, aging, capacity pressure, and eviction are intra-domain.

Neutral/friendly/hostile are identification states, not extra domains.

The same physical truth contact may have a semantic representation in each domain because the records represent different legal knowledge.

RIO priority is a four-track subset of the MEGA/player domain only and is not a third/global track pool.

12.5 Stage-15 internal order

The logical stage-15 order is:

1. project persistent radar/link state and Weapon-owned read-only views into legal stage-15 views;
2. apply radar, jammer, link, damage, and capability effects;
3. generate organic radar observations from truth candidates plus eligible chaff using the approved deterministic detection path;
4. ingest matured MEGA/RED reports; reports denied by current RX state are dropped rather than backlog-buffered;
5. ingest IFF/interrogation/authorized identification evidence;
6. associate/fuse/update semantic tracks inside each isolated domain;
7. generate outbound MEGA/RED reports from newly committed eligible tracks and enqueue for future legal delivery subject to fan-out limits;
8. build passive-IR seeker candidate observations from targets plus eligible flares;
9. build active-radar seeker candidate observations from targets plus eligible chaff;
10. evaluate seeker competition and retention hysteresis;
11. publish final **SeekerAssessment** records to the per-missile mailboxes;
12. publish **WeaponSupportUpdate** records from newly committed valid local fire-control tracks;
13. update RWR emission observations into persistent RWR/threat state.

Helper functions may be rearranged only if observable dependency/order is unchanged and golden/checksum evidence proves equivalence.

12.6 Jammer / link / player-status implementation

When the ownship defensive jammer is active according to the approved Gameplay behavior:

- ownship organic radar operation is denied/blanked;
- MEGA LINK transmit is denied;
- MEGA LINK receive is denied;
- denied in-flight link reports are lost and not replayed later as a backlog;
- existing tracks coast/age by normal rules rather than becoming frozen truth;
- Weapon Support Datalink remains a distinct system and is never implemented as MEGA LINK;

- the `PresentationSnapshot` exposes jammer state, organic-radar-blanked state, and MEGA-denied state so the left status display and RIO/text warning path can explain the consequence immediately.

Exact support-subsystem degradation under jammer/damage is governed by the sensor/weapon/FCS detailed data and later generated tables. Runtime v1 must provide the states/interfaces and evidence hooks without guessing Phase-3 numeric timing.

12.7 IR, RWR, and countermeasure integration

- Passive IR seekers remain electromagnetically silent.
- There is no dedicated incoming-IR MAWS in Version 1.
- Own-heater seeker tone represents the player's own IR seeker.
- Active radar seekers are valid emissions and may produce RWR indication according to the sensor model.
- Chaff/flare may be released whenever otherwise legal, but effectiveness qualification uses achieved positive normal load of at least +3 G in the approved window; commanded G and negative G do not substitute.

The exact window, seeker susceptibility tables, radar tables, and IR-signature tables remain Phase-3 detailed data unless already frozen elsewhere.

13. AIEngine integration

13.1 Ownership and knowledge boundary

AIEngine owns bounded deterministic decision state, held-intent state, doctrine/profile state references, team/formation state, mobility/procedural state, and stage-16 production of **AIIntentFrame** outputs.

It does **not** own:

- sensors or semantic track truth;
- aircraft physical motion;
- FCS/system private state;
- weapon physical/guidance state;
- damage state;
- mission truth outside declared mission views;
- unrestricted global entity truth.

AI never reads global truth.

Legal inputs are explicit read-only views from the owning modules.

13.2 Controller hierarchy

The detailed AI paper owns the doctrinal/state-graph implementation. Runtime defines the integration layers:

HARD SAFETY
MISSION
FORMATION / TEAM
OPERATIONAL / MOBILITY
TACTICAL
MANEUVER / GUIDANCE
CONTROL / INTENT OUTPUT

Interrupt ordering is deterministic. Skill/difficulty changes reaction, precision, doctrine, and decision quality without changing physics or granting hidden knowledge.

Realistic and Novice use the same architecture; Novice modifiers apply to hostile tactical AI only unless a later Gameplay revision explicitly expands scope.

13.3 Stage-16 scheduler

The stage-16 scheduler supports multiple deterministic logical cadences without creating another physical clock.

It shall represent at minimum due-state for:

- reflex/safety evaluation;
- doctrine/tactical evaluation;
- navigation/operational evaluation;
- slower reassessment/coordination services where the AI white paper requires them.

Exact service periods are generated/TBD or white-paper-owned until the applicable implementation gate closes them.

If a service is not due, its valid held intent remains in effect.

13.4 AIIntentFrame

The public intent frame separates at least:

- guidance intent;
- system intent;
- weapon intent;
- sensor intent.

It carries source/owner identity, source tick, validity/lifetime or equivalent stale-intent semantics, and any generation/version fields required to make replay/next-tick application unambiguous.

AI intent is a request/command boundary. It does not directly edit position, velocity, G, control-surface state, sensor detection, missile state, damage, or mission state.

13.5 Legal knowledge views

Generated read-only view interfaces include the minimum concepts required by the AI baseline, such as:

- legal tactical-track view by knowledge domain;
- RWR/threat view;
- visual-contact view;
- own aircraft/system capability view;
- own weapon status and employment projection;
- mission objectives/ROE/posture/task view;
- route/goal/facility view;
- recovery reference;
- facility-clearance view.

A view contains only information the owning system is authorized to expose. Convenience truth fields are prohibited.

13.6 Route, taxi, deck, and ground mobility

Runtime representation is authored/precompiled:

- `RoutePlan` uses bounded legs/continuations/alternates/diverts as required;
- taxi uses authored `TaxiRoute`;
- deck movement uses authored `DeckRoute`/facility state;
- ship routes/patterns are authored;
- ground/mobile-SAM paths are precompiled;
- selecting a ground destination does not invoke general runtime graph search;
- mission interruption preserves bounded suspended-route state and resumes only through authored legal continuation/alternate/divert/recovery paths.

No A*, GOAP, neural model, or general-purpose runtime pathfinding is introduced.

13.7 RIO and player boundary

RIO is AI assistance, not hidden player-aircraft autopilot.

Runtime routes RIO decisions only through approved systems/sensor/countermeasure/jammer/communication intent paths. Player stick, throttle, configuration, and weapon trigger remain owned by the approved player/input/gameplay paths.

RIO-default countermeasure operation is represented explicitly in the intent/interface architecture so the player is not dependent on hidden state. Jammer/CM consequences are exposed through presentation and RIO/text callouts.

13.8 AI resource ledger

R0-D/E and Phase-1 measurement of the AI owner includes:

- resident AI code;
- runtime authoritative AI state;
- per-entity/per-team/per-facility state;
- held-intent state;
- legal-view storage;
- route/doctrine/profile mission-hot data;
- stack/scratch high-water;
- stage-16 scheduling overhead;
- worst legal reflex + doctrine + mobility/operational execution under the approved combined load.

Average AI cost is insufficient.

Production Phase-4 doctrine/tuning does not move backward into R0-D/E simply to make the fixture more realistic.

14. PresentationExtractor and GraphicsEngine integration

14.1 PresentationSnapshot

`PresentationExtractor` owns stage-19 extraction of bounded immutable presentation state. `CoreRuntime` owns stage-21 atomic publication.

A published snapshot is complete and immutable to consumers.

No renderer/audio/UI consumer reaches back into authoritative private arrays for newer values simply because a field was omitted from the snapshot.

The exact snapshot byte size, count, placement, alignment, and field packing are generated/MEASURED and must close in the memory and timing ledgers.

14.2 World construction

World construction is incremental, yieldable, and resumable. Only a complete world buffer is displayed.

A partially constructed world is never exposed as a completed frame.

The renderer consumes the appropriate snapshot/registration generation and may continue constructing a world across multiple authoritative ticks while protected services proceed.

14.3 World registration

Each displayable world buffer carries an atomic `WorldRegistrationRecord` or generated equivalent containing enough identity to keep:

- view/camera generation;
- world-source snapshot/tick;
- terrain visual/query registration;
- spatial cue placement;
- display buffer/generation;

coherent.

The exact record size is generated/MEASURED.

14.4 Occlusion and LOD

Graphics v2.1 owns the detailed hybrid renderer, occlusion, LOD hysteresis, registration-anchor priority, and presentation tiers.

Runtime owns:

- scheduling;
- work-queue bounds;
- workspace ownership;
- DMA submission;
- complete-buffer publication;
- source-generation identity;
- high-water counters;
- protected-service interaction;
- evidence.

Exact face/vertex/span ceilings, LOD distances, RRB production selection, palette details, viewport/resolution, occlusion-envelope resolution, and maximum world age remain MEASURED until the named closure freezes them.

14.5 Protected cockpit and status

Cockpit/HUD/critical warnings are protected independently of outside-world cadence.

The Gameplay-required left status display fields include at least the implementation source for:

- FCS state/status;
- fuel;

- engine state;
- countermeasure inventory;
- jammer ON/OFF state;
- organic-radar-blanked consequence;
- MEGA LINK denied/unavailable consequence where caused by the own jammer.

RIO speech/text receives the same underlying state transition/event rather than inferring it by reading display memory.

15. AudioEngine integration

15.1 Ownership

AudioEngine owns all logical audio arbitration. Only approved PlatformABI wrappers control SID/audio-DMA hardware.

Audio is protected presentation and cannot mutate authoritative simulation, sensors, AI, weapons, mission state, replay state, or checksums.

15.2 Hardware/resource model

Version 1 integrates:

- 4 SIDs / 12 SID voices as the procedural/music foundation;
- 4 pooled PCM channels;
- preferred logical SID domains: SID 1/2 music, SID 3 aircraft/environment, SID 4 warnings/RWR/procedural effects;
- PCM primarily for speech and short high-information transients;
- SID-first fallbacks and text/visual fallbacks required by the audio/gameplay contracts.

The SID allocation is preferred logical ownership, not a reason to bypass the priority scheduler.

15.3 Priority model

The generated audio event model supports P0 through P4 priority classes.

P0/P1 safety/information requirements outrank music, speech, and spectacle. Weapon/catapult/impact PCM does not become P0/P1 merely because it is dramatic.

The scheduler defines deterministic:

- admission;
- voice/channel stealing;
- preemption;

- retrigger;
- ducking;
- loop/cut behavior;
- recovery;
- drop counting;
- speech/text relationship.

15.4 Staging

Audio hardware does not consume Attic addresses directly.

ResourceManager performs bounded Attic-to-chip staging under approved residency and DMA rules. Tactical disk streaming remains prohibited.

Missing optional PCM uses the approved SID/text/visual fallback rather than stalling authoritative work.

15.5 Measured audio values

PCM format/rate/duration, cache split, aggregate rate, exact service clocks, and warning-latency ceilings remain evidence-gated until accepted R0-D/E/F/measured-limits closure.

16. ResourceManager, StorageService, packages, saves, and MissionEngine

16.1 Resource tiers

Attic RAM is a cold/large-resource store, not a second mutable authoritative heap.

Resources required for a live tactical entity/service are either resident or have an approved deterministic bounded staging path before they become required.

A valid simulation tick cannot wait indefinitely for optional Attic or disk content.

16.2 Tactical I/O

Tactical mission packages are preloaded.

Ordinary disk access is inactive during authoritative combat simulation except for a separately approved bounded diagnostic/fault path.

No tactical feature may depend on opportunistic in-sortie disk streaming.

16.3 R0-C disposition

R0-C remains closed. Runtime v1 carries remaining **StorageService** production implementation/acceptance details forward without reopening Charlie.

The reversible post-ROM-reclaim storage-service mechanism, if still required by a production feature, must be resolved before that feature depends on it. It is not assumed merely because R0-C is closed.

16.4 Mission source and compiler

Mission authoring is declarative and compiled offline. JSON5 or its approved successor is host-side authoring input, not a target runtime parser/VM contract.

The mission compiler:

- validates schema/version;
- resolves symbolic resource/entity references;
- selects **AircraftPhysicsClass** per aircraft;
- compiles routes/taxi/deck/ground paths;
- computes bounded pool/resource requirements;
- emits a capacity witness;
- rejects non-droppable capacity violations;
- records required hot resources;
- emits deterministic target packages/chunks;
- emits source/build identity and diagnostics.

The target does not contain a general JSON5 parser, scripting VM, dynamic allocator, or runtime path planner to reproduce host authoring logic.

16.5 Mission truth and AI

MissionEngine owns objectives, scoring, ROE/task state, authored route/task references, and allowed mission posture.

It exposes bounded legal views to AI. It does not grant unrestricted mission/global truth as a convenience API.

16.6 Saves

There is no mid-sortie save in Version 1 unless a later parent revision explicitly adds it.

Persistent campaign/save data is versioned, validated, bounded, and recoverable according to Gameplay/Storage requirements. Save migration and corrupt-media behavior are explicit; unknown versions do not silently reinterpret bytes.

17. Replay, checksum, faults, diagnostics, and evidence

17.1 Canonical deterministic state

All authoritative state that can affect future simulation behavior participates in the canonical deterministic model, including relevant:

- entity/pool state;
- aircraft/system/flight state;
- weapon/damage state;
- SensorTrack and SensorHandoff state;
- AI state and held intents;
- mission/objective state;
- RNG streams;
- pending deterministic commands/events/mailboxes.

Presentation-only buffers, debug traces, and non-authoritative instrumentation are excluded unless their inclusion is required to detect corruption and is defined not to affect behavior.

17.2 Checksum

Stage 20 calculates the canonical checksum over a generated, versioned authoritative-state traversal order.

The traversal is independent of:

- pointer address;
- source-file iteration order;
- display order;
- hash-table order;
- renderer cadence;
- debug log order.

The generated checksum map identifies every included field/owner.

17.3 Replay

Replay/golden fixture input records enough external and nondeterministic boundary state to reproduce authoritative results for the test class, including approved input and RNG seed/stream identity.

Replay never records a presentation result and feeds it back as simulation truth.

17.4 Fault model

Each bounded queue/pool/resource/service has declared overflow/failure behavior. Fault injection covers, as applicable:

- pool exhaustion;
- queue overflow;
- stale handle/generation;
- stale mailbox/handoff;
- memory canary violation;
- stack high-water/guard;
- DMA deadline/contention fault;
- presentation backlog/world-age fault;
- audio channel/cache/staging contention;
- storage/media fault;
- resource absence/fallback;
- sensor/link denied/drop behavior;
- deterministic timing fault.

A fault never grants permission to change the authoritative clock or silently drop a required authoritative entity.

17.5 Evidence identity

Every major gate/evidence set records at minimum:

- source commit;
- exact approved specification identities;
- generated schema/registry identities;
- toolchain lock identity;
- target build profile and flags;
- target binary/D81/package hash;
- host OS/tool identities needed for reproduction;
- Xemu identity/configuration where used;
- physical MEGA65/core/ROM/system/video-mode identity where required;
- maps/symbols/listings/disassembly where applicable;
- tests/golden vectors;
- memory/code/stack/DMA/cycle/latency/high-water measurements;
- pass/fail and waiver identity;
- human acceptance where required.

Xemu is not physical-MEGA65 evidence when the gate explicitly requires hardware confirmation.

18. Repository, generated artifacts, and AI-assisted engineering

18.1 Repository is the implementation control plane

The active design corpus is deliberately small. The repository may contain many legitimate engineering-control artifacts without promoting them to design-document authority.

Required logical artifacts include at least:

```
spec/approved-spec-set.json
interfaces/f65_interfaces.json5
interfaces/f65_numeric_registry.json5
interfaces/f65_platform_abi.json5
memory/f65_memory_ledger.json5
assets/f65_asset_manifest.json5
missions/f65_mission_schema.json5
tests/f65_acceptance_catalog.json5
tests/f65_evidence_schema.json5
toolchain/f65_toolchain.lock.json
```

Equivalent approved paths/names are acceptable. The important contract is ownership, generation, traceability, and one source of truth.

The repository also maintains:

- machine-readable module status/source ownership;
- generated human-readable status board;
- path/module ownership rules;
- diff-scope validation;
- task/plan admission records;
- build/evidence indices.

18.2 Agent/task admission

Every AI-assisted implementation task names:

- requirement IDs;
- approved decision/correction IDs if applicable;
- owning module and authorized paths;
- input/output interface versions;
- state owner and update stage/cadence;
- memory/code/cycle/DMA budget class or MEASURED evidence obligation;
- data/asset/package IDs;
- normal behavior;
- edge behavior;
- overflow/fault behavior;

- host oracle/golden vectors;
- target/Xemu/hardware acceptance IDs;
- expected evidence artifacts;
- whether human review is required.

An automated diff-scope check rejects edits outside declared scope unless the task is explicitly widened.

18.3 Allowed autonomous work

Within an approved task and interface, an AI coding agent may perform:

- local C implementation;
- approved assembly/platform wrappers;
- generators and schemas;
- host reference models and vectors;
- deterministic fixtures and validators;
- instrumentation/evidence capture;
- build/toolchain work;
- behavior-preserving refactoring;
- measurement-driven optimization;
- required same-change documentation/status updates.

18.4 Stop/escalate conditions

An agent must stop the affected work instead of inventing semantics when the task requires:

- changing Main Concept architecture;
- new player-visible gameplay;
- changing public ABI/state ownership without approval;
- changing memory map, hard pool capacity, or tick order;
- selecting a TBD/TARGET/MEASURED production value without the required owner/evidence;
- relying on undocumented MEGA65 behavior;
- bypassing MemoryAccessABI/PlatformABI/CoreRuntime ownership;
- introducing a second physical clock;
- giving AI hidden/global truth;
- inventing mission/campaign content, control feel, art/audio content, or an acceptance waiver.

Independent reversible work may continue if it does not depend on the blocked decision.

18.5 Handoff contents

Every completed AI-assisted change includes:

- diff/commit identity;

- generated artifact report;
 - tests;
 - golden-vector status;
 - size/cycle/stack/DMA delta where applicable;
 - memory-ledger delta;
 - invariant/requirement references;
 - evidence links/IDs;
 - unresolved measured/TBD items;
 - explicit statement of whether physical hardware evidence is still required.
-

19. R0-D, R0-E, R0-F, and measured-limits closure

19.1 R0-D - current protected-workload calibration

R0-D uses the historical 530,000-clock protected non-render workload as a calibration baseline and adds the measurement hooks required by the current frozen/high-level architecture and accepted detailed subsystem direction.

R0-D includes, as applicable:

- total/stage/protected-service timing;
- snapshot publication timing;
- memory/MAP/DMA/input/storage timing;
- world source-generation and displayed-world-age counters;
- renderer pool/occlusion/registration/LOD high-water;
- protected compositor and graphics DMA accounting;
- AudioEngine service/channel/preemption/staging/P0/P1 counters;
- AI owner code/state/route/doctrine/held-intent measurement hooks;
- bounded stage-16 scheduler/`AIIntentFrame` fixtures proving next-tick causality;
- reserve accounting.

R0-D does not become a second renderer implementation phase, production flight-model phase, radar-table phase, or Phase-4 doctrine phase.

19.2 R0-E - combined Xemu integration

R0-E combines the final pre-Phase-1 proof configuration in Xemu, including:

- independent authoritative simulation and presentation/audio/input service timing;
- complete snapshot publication/consumption;
- memory/resource staging;
- renderer/world registration/occlusion/LOD/complete-buffer behavior;

- cockpit/HUD/protected presentation;
- audio warning/music/SFX/speech contention;
- input edges;
- DMA/IRQ contention;
- storage/resource behavior;
- deterministic faults and shedding;
- bounded AI interface/scheduler/held-intent/legal-knowledge fixtures;
- required combined entity/presentation load.

Later Phase-2/3/4 models may remain bounded proxies/fixtures during R0-E. The fixture must prove integration shape and budgets without smuggling production tuning backward.

19.3 R0-F - physical MEGA65 confirmation

R0-F runs the identified accepted R0-E configuration on physical MEGA65 hardware and performs the required timing/phase sweeps.

The evidence identifies the exact hardware/core/ROM/system/video/storage/input configuration.

A materially changed configuration after a hardware failure returns to the appropriate upstream validation and creates a new identified E-to-F chain.

19.4 Measured-limits closure

After accepted R0-F, the measured-limits process freezes only values supported by evidence, including as applicable:

- protected-service ceilings;
- per-stage/combined workload budgets;
- warning/input latency ceilings;
- renderer/display tier;
- completed-world/world-age limits;
- DMA batch/latency limits;
- RRB/VIC-IV composition selection;
- audio service/channel/cache/latency ceilings;
- code/data/stack/reserve limits;
- resource/scene/presentation pool/work-queue limits still designated MEASURED;
- generated record placements whose width depends on measured implementation choice.

Measured limits are pinned generated/configuration artifacts, not a new design-authority document.

20. Phase 1 - integrated 65Aero engine harness

20.1 Purpose

Phase 1 builds and proves the production-shaped 65Aero backend before the detailed Phase-2 flight, Phase-3 radar/weapon, and Phase-4 tactical implementations are filled in.

The goal is a real engine skeleton with final ownership, timing, interfaces, memory accounting, packages, diagnostics, and evidence, using bounded proxies where later subsystem behavior is not yet authorized.

20.2 Mandatory Phase-1 implementation scope

Phase 1 includes production-shaped:

- CoreRuntime and 21-stage dispatcher;
- PlatformABI and MemoryAccessABI boundaries;
- fixed entity pools/lifecycle;
- generated public interface system;
- generated numeric and platform registries;
- memory/code/stack/DMA ledgers;
- command/event infrastructure;
- InputCommandFrame path;
- AircraftPhysicsClass mission-load representation;
- SIX_DOF and KINEMATIC integration stubs/proxies sufficient to prove ownership/class timing without claiming Phase-2 physics completion;
- sensor/weapon handoff schema including both track domains and SensorHandoffState without claiming Phase-3 radar tuning;
- stage-16 AI scheduler/held-intent/legal-view schema without claiming Phase-4 doctrine completion;
- MissionEngine/package/capacity witness;
- ResourceManager/StorageService production boundary;
- PresentationExtractor and PresentationSnapshot publication;
- GraphicsEngine work scheduling/world registration/protected presentation integration;
- AudioEngine scheduler/platform boundary;
- replay/checksum/fault/diagnostic/evidence system;
- one-command reproducible build/test/evidence path.

20.3 Phase-1 hard memory gate

Phase 1 cannot pass until the generated \$010000-\$017FFF active-simulation owner ledger closes with the actual Version-1 public records and state admitted for the harness.

The proof explicitly includes both SensorTrack domains, SensorHandoffState accounting, system state, command/intent state, and AI state/held intent.

No unresolved overflow is hidden in measured-limits reserve.

20.4 Phase-1 timing gate

Phase-1 timing evidence uses the required nine-aircraft overlap and the applicable combined-load fixtures.

At minimum it proves:

- exact 100 Hz authoritative tick under the identified harness;
- deterministic 21-stage order;
- no same-tick Sensor->Weapon or AI->physical feedback;
- bounded stage-15/16 proxy/fixture cost;
- protected presentation/audio/input deadlines from measured limits;
- complete world buffer semantics and world-age instrumentation;
- baseline 6 SIX_DOF + 3 KINEMATIC aircraft class admission;
- separately identified SIX_DOF AIC harness variant admission or explicit failure/evidence if the variant is still conditional;
- deterministic checksums/replay under repeated identical runs;
- pool/queue high-water below hard capacity for all required non-droppable classes.

20.5 Phase-1 acceptance

Phase 1 is PASS only when:

1. the required generated interfaces are current and pinned;
2. memory/code/stack/runtime-support/DMA ledgers pass;
3. the 21-stage harness passes deterministic replay/checksum tests;
4. the baseline combined-load harness passes its R0/measured timing bars;
5. all public ownership boundaries are enforced in code/tests;
6. mission capacity analysis proves the harness package;
7. protected input/HUD/warning/audio and world-buffer rules pass;
8. Xemu evidence passes;
9. physical-hardware requirements inherited from measured-limits/R0-F remain satisfied by the Phase-1 configuration or are rerun if materially invalidated;
10. no Phase-2/3/4 implementation is falsely claimed complete merely because a proxy exists.

21. Downstream plug-in boundaries

21.1 Phase 2 - flight and aircraft systems

Phase 2 fills the production Physics v3.3 and aircraft-system behavior behind the Phase-1 interfaces.

It closes:

- coefficient/mass/inertia/engine/atmosphere tables;
- FCS gains and degraded-FCS mapping;
- actuator/hydraulic/system dependencies;
- AUGMENTED, AUG_OFF, DIRECT_ONLY, ADLC, autothrottle behavior;
- full F-65A SIX_DOF implementation and oracle acceptance;
- final approved KINEMATIC aircraft implementation details;
- carrier/airfield physical integration required for Phase-2 acceptance.

It does not rewrite the Runtime ownership model.

21.2 Phase 3 - sensors, weapons, countermeasures, damage

Phase 3 fills the Radar/Sensor and Weapon/Damage behavior behind the Phase-1 schemas and Phase-2 physical state.

It closes, among other things:

- radar detection/scan/clutter/notch tables;
- track association/quality/aging values;
- tactical-link cadence/latency/range/fan-out/queue values;
- seeker/IR/countermeasure tables;
- weapon support/autonomy timing;
- missile/gun/damage detailed behavior;
- exact **SensorHandoffState** field semantics/supersession where still gated;
- required sensor/weapon host oracles and golden vectors.

21.3 Phase 4 - tactical AI and combat slice

Phase 4 fills the human-reviewed AI doctrine/state graphs, detailed cadences, utility weights, maneuver choices, proficiency tuning, IADS, mobility/procedural behavior, and tactical content behind the Phase-1 AI interface.

The Technical Combat Slice precedes Midnight Spear as the production tactical validation step.

Novice remains hostile-only tactical modification unless Gameplay changes.

21.4 Phase 5 - campaign and release

Phase 5 completes the campaign/content manifests, compatibility, optimization, save migration/recovery, release evidence, and human acceptance without relaxing the frozen engine contracts.

22. Gameplay acceptance scenes mapped to Runtime evidence

The Runtime evidence catalog shall contain stable IDs for the six frozen Gameplay acceptance scenes plus the separate training/usability TARGET evaluation.

22.1 Ordinary cruise

Runtime evidence proves:

- 100 Hz authoritative simulation;
- stable SIX_DOF path when Phase 2 is present or the production-shaped Phase-1 proxy before then;
- protected HUD/cockpit/input/audio;
- complete world buffers only;
- displayed-world age/cadence counters;
- approximately 30 Hz completed-world TARGET evaluated in the applicable measured configuration.

22.2 Combined fleet fight

Runtime evidence proves the required combined overlap:

- 9 aircraft, baseline 6 SIX_DOF + 3 KINEMATIC;
- 16 guided missiles;
- 24 gun projectile groups;
- 48 decoys;
- both tactical knowledge domains;
- AI/RIO work appropriate to the current phase;
- damage/events;
- protected HUD/audio;
- world rendering;
- exact 100 Hz simulation;
- approximately 25 Hz world TARGET;
- sustained below-20-Hz world cadence treated as failure;
- memory/queue/pool/timing high-water recorded.

22.3 AUG_OFF handling

Runtime evidence proves the state/command boundary:

- FCSMode=AUG_OFF is not a hidden rate-command law;
- FCSStatus remains independent;
- ADLC transition occurs correctly;
- pilot trim and temporary transfer bias are distinct;
- no mode-switch impulse caused solely by state transition;

- autothrottle remains separately represented.

Detailed aircraft-feel acceptance remains Phase-2/human physics acceptance.

22.4 Carrier Case I / recovery registration

Runtime evidence proves:

- carrier authoritative motion/contact remains at 100 Hz;
- displayed deck/world and spatial cues share the displayed `WorldRegistrationRecord` generation;
- HUD/cockpit remains protected while world construction is late;
- arrestment/contact uses authoritative geometry, not displayed interpolation.

22.5 Warning/audio contention

Runtime evidence proves:

- P0/P1 arbitration and latency counters;
- SID/PCM high-water;
- voice/channel stealing and fallback;
- warnings remain intelligible/available under music/SFX/speech contention;
- audio DMA/resource staging cannot block authoritative simulation.

Human audio quality remains a separate acceptance step.

22.6 Sensor/weapon/defense scenario

Runtime evidence proves the architecture-level causality and visibility contracts:

- two isolated 24-track domains;
- offboard data cannot create ownship weapon-quality/support by itself;
- jammer blanks organic radar and denies MEGA TX/RX;
- denied reports are dropped, not replayed;
- `SensorHandoffState` N->N+1 causality;
- +3 G actual-load CM qualification path;
- no incoming IR MAWS;
- own-heater audio remains own-seeker state;
- jammer/MEGA/radar consequences reach Gameplay status/RIO presentation.

Detailed radar/seeker probabilities and weapon outcomes remain Phase-3 data.

22.7 PAL/NTSC authoritative equivalence

Runtime evidence proves that supported PAL/NTSC presentation differences do not change:

- authoritative 100 Hz simulation;
- deterministic event order;
- sensor/weapon/AI causality;
- RNG sequence;
- authoritative checksum for an equivalent test input/seed.

Presentation cadence may differ only within the supported measured presentation contract.

22.8 Training/usability TARGETS

The first-time completion-in-two-attempts and 3-of-5 carrier-trap criteria remain Gameplay TARGETS pending implemented Operations 1-2 and human review.

Runtime provides telemetry/evidence hooks but does not make those human-use metrics Phase-1 or architecture MUSTs.

23. Required Version-1 generated freeze package

The following are mandatory for Runtime v1 final freeze, using these logical names or approved equivalents:

Artifact	Required role
<code>spec/approved-spec-set.json</code>	Exact active Main, Gameplay, Runtime candidate/final, white-paper, decision, and evidence identities.
<code>interfaces/f65_interfaces.json5</code>	Canonical public record/API semantics, fields, widths, offsets, ownership, lifetime, stage/cadence.
<code>interfaces/f65_numeric_registry.json5</code>	Fixed-point formats, enum mappings, ranges, rounding/saturation, generated constants.
<code>interfaces/f65_platform_abi.json5</code>	C/assembly/PlatformABI entry points, clobbers, map/base-page/stack/IRQ assumptions.
<code>memory/f65_memory_ledger.json5</code>	Physical ranges, owners, record arrays, alignment, stack, code/data, reserve, overlap proof.
<code>assets/f65_asset_manifest.json5</code>	Resource IDs, classes, resident/fallback/staging policy, converted sizes and budgets.

Artifact	Required role
missions/f65_mission_schema.json5	Declarative source/compiled mission contract, physics-class selection, routes and capacity inputs.
tests/f65_acceptance_catalog.json5	Stable test/scene/gate IDs and required platform/evidence levels.
tests/f65_evidence_schema.json5	Machine-readable evidence records, counters, units, build identity and pass/fail.
toolchain/f65_toolchain.lock.json	Exact compiler/assembler/linker/runtime/Xemu/host/target identity.

Additionally required:

- generated C headers;
- generated assembly include files;
- generated Java bindings/reference records;
- static-assert compilation test;
- module ownership/status source;
- generated status board;
- diff-scope validator;
- clean/stale-generated-output check.

Freeze gate: The Runtime v1 prose is now **human-approved as the candidate design**. That approval authorizes generation and closure work against this constitution; it does not make the document FINAL. **FINAL / HUMAN-REVIEWED Runtime v1 requires the Version-1 generated freeze package to be pinned and internally consistent.** The first generated control artifact shall pin the exact approved Main v1.6 and Gameplay v1 filenames/hashes and this approved-candidate Runtime identity; repository aliases or duplicate upload suffixes such as (1) are not specification identity.

24. Current MEASURED / TBD register

This section prevents implementation from filling unresolved values by taste.

ID	Item	Class	Closure owner/gate
RT-M-001	Final per-stage CPU ceilings and protected-service budgets	MEASURED	R0-D/E/F + measured limits
RT-M-002	Exact warning/input latency ceilings	MEASURED	R0-D/E/F + measured limits
RT-M-003	Maximum displayed-world age	MEASURED	R0-D/E/F + measured limits
RT-M-004	Final renderer/work-queue/LOD/occlusion ceilings still marked R0-gated	MEASURED	Graphics + R0-D/E/F
RT-M-005	Final PCM format/rate/cache/aggregate/latency limits	MEASURED	Audio + R0-D/E/F
RT-M-006	Exact PresentationSnapshot width/count/placement	MEASURED/generat	Runtime interface + memory + R0 timing closure
RT-M-007	Exact WorldRegistrationRecord width/placement	MEASURED/generat	Runtime/Graphics + memory closure
RT-M-008	Exact SensorTrack width/alignment and total two-domain charge	MEASURED/generat	Runtime/Sensor schema + active-sim closure
RT-M-009	Exact SensorHandoffState widths, link queue cap, supersession and total charge	TBD/MEASURED generated	Runtime/Sensor Phase-3 schema; pre-Phase-3 interfaces must reserve/represent it without guessing
RT-M-010	Exact AIIntentFrame intent/held-intent widths	TBD/MEASURED subgenerated	Runtime + AI generated interface; R0-D/E owner measurement

ID	Item	Class	Closure owner/gate
RT-M-011	Exact AI logical service cadences not already fixed by approved detailed engineering	TBD	AI Phase-4 implementation gate; R0 fixtures remain bounded proxies
RT-M-012	Degraded-FCS mapping table	TBD	Physics/Phase 2 generated data
RT-M-013	Exact KINEMATIC aircraft approximation details	TBD/MEASURED	Physics/AI/Phase 2; Runtime owns bound-ary/evidence
RT-M-014	Radar/link/seeker/IR/ROM numeric tables	TBD	Phase 3
RT-M-015	Production doctrine weights/proficiency/tactical tuning	TBD	Phase 4
RT-M-016	Remaining StorageService post-ROM-reclaim production mechanism if required	TBD	StorageService before dependent Phase-1/production feature
RT-M-017	Ops 3-10/campaign detailed content	TBD	Phase 5 / human authored content

No listed item blocks independent work whose interface does not require the unresolved value.

25. Draft-0.2 disposition and Version-1 change summary

25.1 Preserved from Draft 0.2

Version 1 preserves the following engineering direction unless a frozen parent explicitly changed it:

- C-primary target implementation;

- selective evidence-driven 45GS02 assembly;
- Java host tooling/oracles/generators;
- deterministic restricted C;
- narrow generated PlatformABI;
- MemoryAccessABI protection;
- sacred CoreRuntime and module isolation;
- fixed pools and deterministic lifecycle;
- one 100 Hz authoritative physical clock;
- declarative offline mission toolchain;
- bounded resources and no tactical disk streaming;
- generated interfaces and asset/package tooling;
- reproducible build profiles and evidence;
- replay/checksum/fault infrastructure;
- repository-scoped AI-assisted development discipline.

25.2 Replaced from Draft 0.2

Version 1 replaces or retires:

- Revision 1.5.1 as active parent;
- Gameplay 0.2 candidate as active companion;
- Technical Alignment / Read-First as active authority;
- the old R0-A-next-work posture;
- any implication R0-A/B/C can be reopened simply to incorporate later white-paper detail;
- the old single 24-track semantic-track memory interpretation;
- placeholder/generic SensorAndTrack handoff semantics;
- generic AI placeholder architecture;
- stale **Assisted/Manual** FCS terminology;
- stale **SimulationSnapshot** terminology;
- any pre-R0 planning memory or cycle allocation treated as a frozen shipping value;
- any display/presentation model that can show a partially constructed world;
- any AI motion path that becomes a second physical authority;
- any MEGA LINK implementation that substitutes for weapon-support datalink or grants shared truth.

25.3 New Version-1 obligations

Version 1 explicitly adds:

- frozen Main v1.6 / Gameplay v1 authority;
- current R0-D/E/F and measured-limits program;
- two domain-isolated 24-track tactical knowledge stores;
- fixed **SensorHandoffState** ownership and N->N+1 sensor/weapon hand-off;

- exact Stage-15 logical order;
 - **AUGMENTED** / **AUG_OFF** / **FCSStatus** / **ADLC** / independent autothrottle implementation boundary;
 - **SIX_DOF** / **KINEMATIC** mission-compiled aircraft class with no second clock;
 - required 6+3 nine-aircraft baseline and separate **SIX_DOF-AIC** variant;
 - stage-16 AI scheduler, legal knowledge, **AIIntentFrame**, held-intent and route/facility interface obligations;
 - AI owner code/state/cycle measurement;
 - world registration, complete-buffer publication, displayed-world age, and 30/25/20 presentation bars;
 - final Audio v1 SID/PCM priority integration and evidence counters;
 - Gameplay-required jammer/organic-radar/MEGA-denial status path;
 - explicit generated freeze package as a condition of Runtime v1 finalization;
 - acceptance evidence mapped to the frozen Gameplay scenes.
-

26. Human-review checklist for this candidate

Before approving Runtime v1 as FINAL, human review should answer **YES** to each item:

1. Does the document implement Main v1.6 without creating new architecture?
2. Does it implement Gameplay v1 without silently changing player behavior?
3. Are all five white-paper domains integrated without duplicating their tuning tables?
4. Is the 100 Hz / 21-stage / next-tick causality model exact and unambiguous?
5. Are **SIX_DOF** and **KINEMATIC** explicitly one-clock, FlightDynamics-owned classes?
6. Are the 6+3 baseline and separate **SIX_DOF-AIC** variant represented correctly?
7. Are two 24-track domains and **SensorHandoffState** correctly owned/accounted?
8. Is the stage-15 logical order correct?
9. Is AI legal-knowledge-only, stage-16, held-intent, fixed-capacity, and pathfinding-free?
10. Are **AUGMENTED**, **AUG_OFF**, **FCSStatus**, **ADLC**, autothrottle, trim, and transfer bias correctly separated?
11. Is the presentation model complete-buffer-only with explicit world registration and world-age measurement?
12. Are 30 Hz cruise and 25 Hz combat kept as **TARGETS** and sustained <20 Hz treated as failure?
13. Is Audio v1 integrated with P0/P1 protection and four pooled PCM chan-

- nels without freezing unmeasured format/latency values?
14. Are R0-A/B/C preserved as closed while current work begins at R0-D?
 15. Is the historical 530,000-clock value clearly a Delta calibration fixture rather than a shipping budget?
 16. Is the \$010000-\$017FFF active-simulation closure made a hard generated-ledger gate?
 17. Is measured-limits reserve protected from silent baseline consumption?
 18. Are unresolved values visibly MEASURED/TBD rather than guessed?
 19. Are the six frozen Gameplay acceptance scenes mapped into Runtime evidence?
 20. Are the generated interface/ledger/toolchain/evidence artifacts mandatory before FINAL freeze?

If all twenty are YES and the generated freeze package is pinned and internally consistent, this candidate is suitable for final human-review disposition.

Appendix A - controlled rewrite inputs

This candidate was drafted against the current project corpus with the following authority posture:

Active frozen parents

- F65_Main_Concept_v1.6_FINAL_HUMAN_REVIEWED.md - FINAL / HUMAN-REVIEWED active master baseline.
- F65_Gameplay_and_Simulation_Supplement_v1_FINAL_HUMAN_REVIEWED.md - FINAL / HUMAN-REVIEWED player-facing baseline.

Transitional source being superseded

- F-65 Engine Runtime and Toolchain Design Supplement Draft 0.2.md - retained as engineering provenance and implementation source where consistent with the frozen parents.

Detailed subsystem engineering inputs

- Flight Physics and Simulation Engineering White Paper v3.3 - frozen governing detailed physics/FCS model.
- Graphics Engineering White Paper v2.1 - detailed graphics engineering input under the current Main/Gameplay adoption and remaining R0-D/E/F measurement gates.
- Audio, Sound Effects and Music Engineering White Paper v1.0 - final human-reviewed audio engineering baseline.

- Radar, Sensors and Track Engineering White Paper v1.0 / SensorAndTrackEngine Phase-3 v1.0 - human-reviewed sensor/track engineering baseline.
- AI Behavior and Decision Architecture White Paper v1.0 - final human-reviewed AI engineering baseline.

Historical architecture/alignment/GAD sources are provenance only and cannot override the active parents.

Appendix B - first generated-interface work after candidate approval

The first implementation-facing work after prose approval should be generated-interface closure, not gameplay coding.

Recommended sequence:

1. pin the approved Main v1.6 and Gameplay v1 identities in **approved-spec-set**;
2. import/normalize the proven R0-A PlatformABI and toolchain lock rather than recreating them from prose;
3. generate the common identity/command/event/input records;
4. generate **AircraftPhysicsClass** and FCS state enums/records;
5. generate **SensorTrack**, two-domain container semantics, **SensorHandoffState**, and sensor/weapon views/mailboxes;
6. generate **AIIntentFrame**, sub-intents, held-intent state, legal views, routes and facility/recovery records;
7. generate **PresentationSnapshot** and **WorldRegistrationRecord** proof forms;
8. run the full \$010000-\$017FFF owner-ledger closure;
9. add R0-D counters missing for Graphics/Audio/AI/Sensor handoff/world age;
10. build the bounded R0-E/Phase-1 proxy fixtures against those exact interfaces.

No Phase-2 physics coefficients, Phase-3 radar tables, Phase-4 doctrine weights, or Phase-5 campaign content should be invented during this interface-closure sequence.

End of F65 65Aero Engine Runtime and Technical Supplement v1 - APPROVED CANDIDATE DESIGN - NOT FINAL